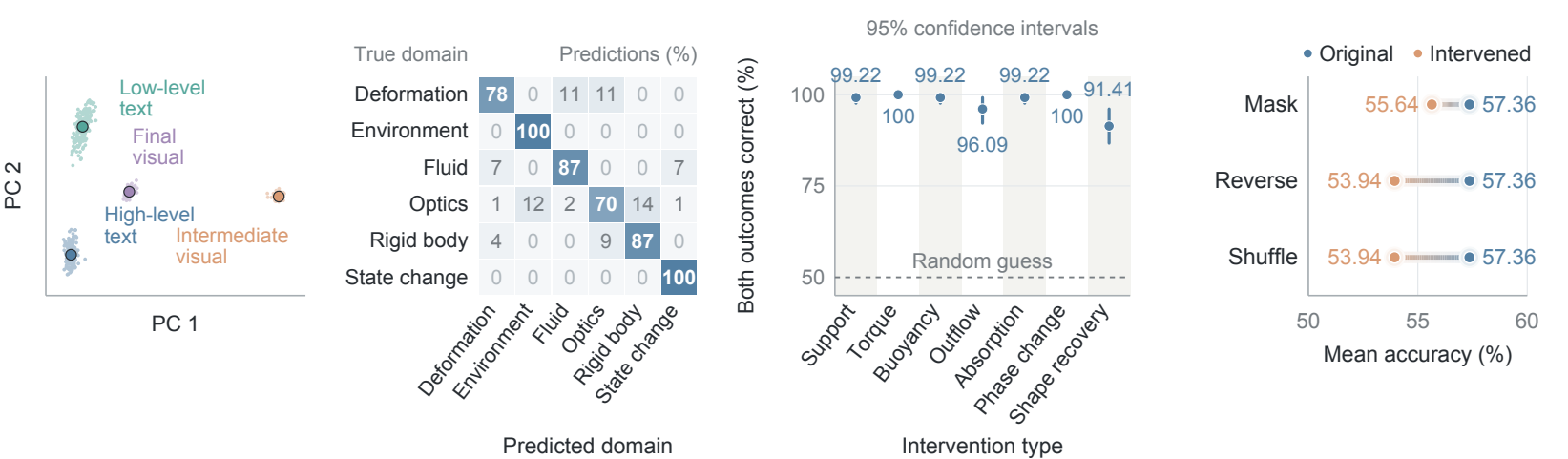



**(a)** Token-role structure

**(b)** Physical-domain linear probe

**(c)** Physical-intervention prediction

**(d)** Visual-token perturbation